indeed favourable for the growth of the design industry, there are a few impediments that need to be removed for the industry to live up to the expectations.

Over the last couple of decades, the Indian electronic design industry has gradually shifted focus from manufacturing towards assembly. Earlier, the entire production and assembly of products was done in India. However, now semi-finished items are imported from China and only the final assembly is done in India. For example, mobile phones and smart TVs are designed in China and assembled in India. Only a small percentage of design for local brands is done in India.

One of the prime reasons behind this trend is the requirement of significant investment for R&D in a sector where margins are already thin. Pressed for margins, firms prefer to shop for products that are suitable for the Indian market and arrange for imports of semi-knocked-down kits back to India for assembly.

Other factors impeding the industry growth include:

1. SMEs and startups lack experience in complex system design and are completely unaware of outsourcing opportunities in both design and manufacturing.

2. Design (CAD) tools are expensive, while open source tools are not of acceptable quality. As engineering design is capital-intensive, educational institutes with relevant CAD and T&M equipment could offer their labs on hire, helping startups and SMEs to develop products at low cost.

3. There is dearth of hardware accelerators supporting board design and FPGAs/EPLDs.

4. Governance/policy related challenges:

(i) Prototype ecosystem is yet to mature, leading to a lot of time wastage in customs and other processes.

(ii) Duty-free components are still being charged at 28 per cent GST.

5. While there is a steady supply of graduates in the market, there is still a significant gap between skill sets required by the industry and skill sets acquired by graduates through higher education institutes. Effective collaboration programmes between the industry and institutes can go a long way in developing more employable candidates for the industry.

Supportive policies

The government of India is determined to resolve these issues by developing policies and frameworks that can catalyse growth in the design sector. Initiatives like Make in India and Digital India, and impetus to existing schemes like the modified special incentive package scheme (M-SIPS) and electronic development fund (EDF) will have a significant impact in driving this change.

The industry has responded positively to the government's support. As many as 159 new ESDM units were established in 2015 itself. According to industry experts, while Make in India is an initiative in the right direction, the agenda could also include 'Make for India' with focus on Indian priorities.

Other supportive measures taken by the Indian government include:

1. 100 per cent FDI allowed in the electronics hardware manufacturing sector under automatic route

2. Duty relaxation and schemes such as EPCG, EHTP and SEZ to provide tax sops; duty exemption on equipment required for setting up semiconductor plants

3. National Policy on Electronics (2012) and setting up of National Electronics Mission EFY

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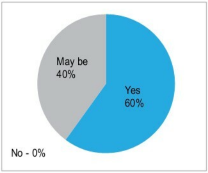

Fig. 3: India's potential to become the design house for the global electronics industry

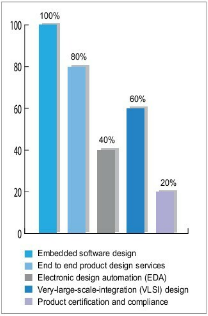

Fig. 4: Design services in demand

Types of electronics design companies

The Indian electronics design industry consists of:

1. Independent design houses offering design services for the world market more under the services business model (driven by IoT, automation, analytics, etc) and designing India-specific products (especially with the IoT and automation driving efficiencies for manufacturing, healthcare and hospitality segments)

2. Captive design houses of global electronics firms

In addition to that, there are EDA tool providers and original design manufacturers catering to the market.